

Multimodal Learning for Medicine & Healthcare: The Challenges and Opportunities (IN2107)

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Chair for Computational Imaging and AI in Medicine
Winter Semester 2026/2027

About Us



Chair for Computational Imaging and AI in Medicine
Prof. Dr. Julia Schnabel

About Us



Dr. Laura Daza

- Medical Image segmentation
- Foundation models
- Multimodal learning

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Marta Hasny

- Foundation models
- Multimodal learning
- Cardiology

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Jun Li

- Large Vision Language Model
- Abnormality Grounding
- Multimodal Learning

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Dr. Cristina González

- Medical Image segmentation
- Foundation models
- Multimodal learning

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1) Organization

Preliminary Schedule (Subject to Change)

Week	Session
Week 1	Introduction to the Seminar
Week 2	How to read a paper and do poster presentations
Week 3	Self-supervised Learning
Week 4	Challenges of Multimodal Learning in Medicine
Week 5	Student Presentations
Week 6	Student Presentations
Week 7	Student Presentations
Week 8	Student Presentations
Week 9	Invited Talk
Week 10	Multimodal Foundation Models
Week 11	Student Presentations
Week 12	Student Presentations
Week 14	Invited Talk
Week 15	Poster Session

Where?

5607.01.023 (Seminarraum)
MI building

When?

Mondays: 14:00-16:00

Deliverables & Grading

Oral presentations

- 30 minutes presentation + 10 minutes questions
 - About your selected paper + small review of the topic
- Presentation date depends on the topic

Poster presentations

- Scientific poster
 - About your selected paper only
- 5 minutes presentation

Paper selection

- We will provide a list of papers
- We will assign the papers based on your submitted selections
- Papers on both multimodal methodologies and medical applications

Grading

Oral + poster presentations + attendance + participation



Poster session at end of semester

Paper Topics from Previous Semesters

- Multimodal Fusion
- Addressing the Challenges of Multimodal Learning
- Multimodal Pretraining
- Multimodal Medical Foundation Models
- Multimodal Generalization & Representation

We try to select recent papers while providing the background to better understand them in the lectures. That way you can have an overview of what is going on in the field right now 😊

2) Multimodal learning

What will the seminar be about?

Multimodal Learning

Vision-Language Models

(1) Contrastive pre-training

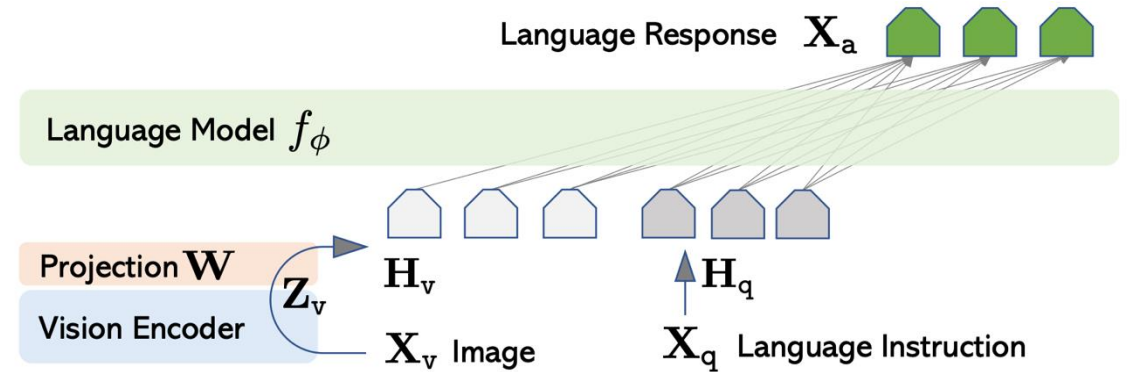
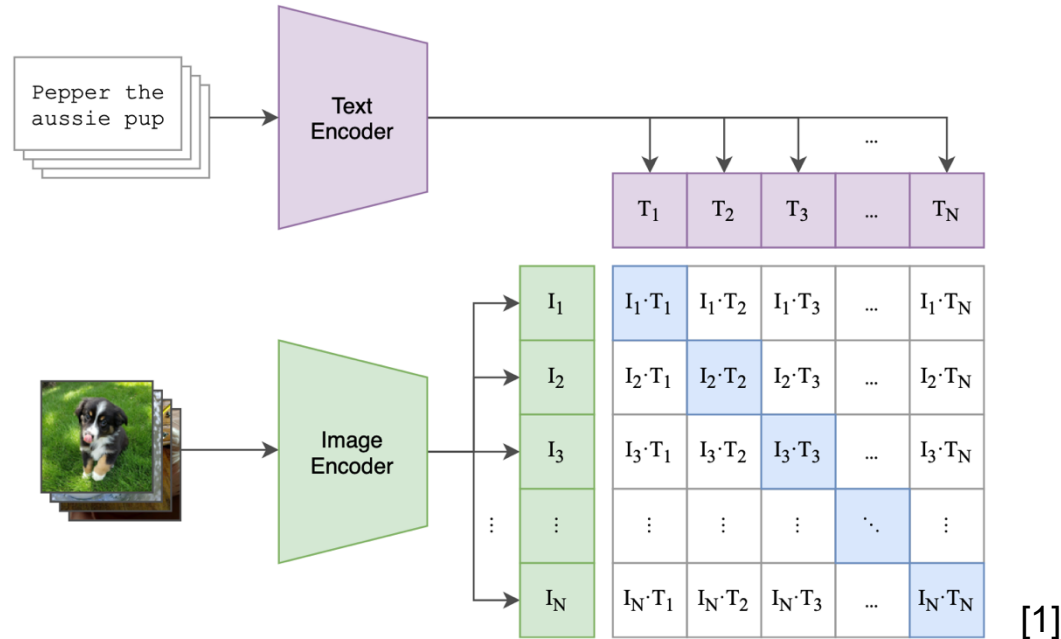


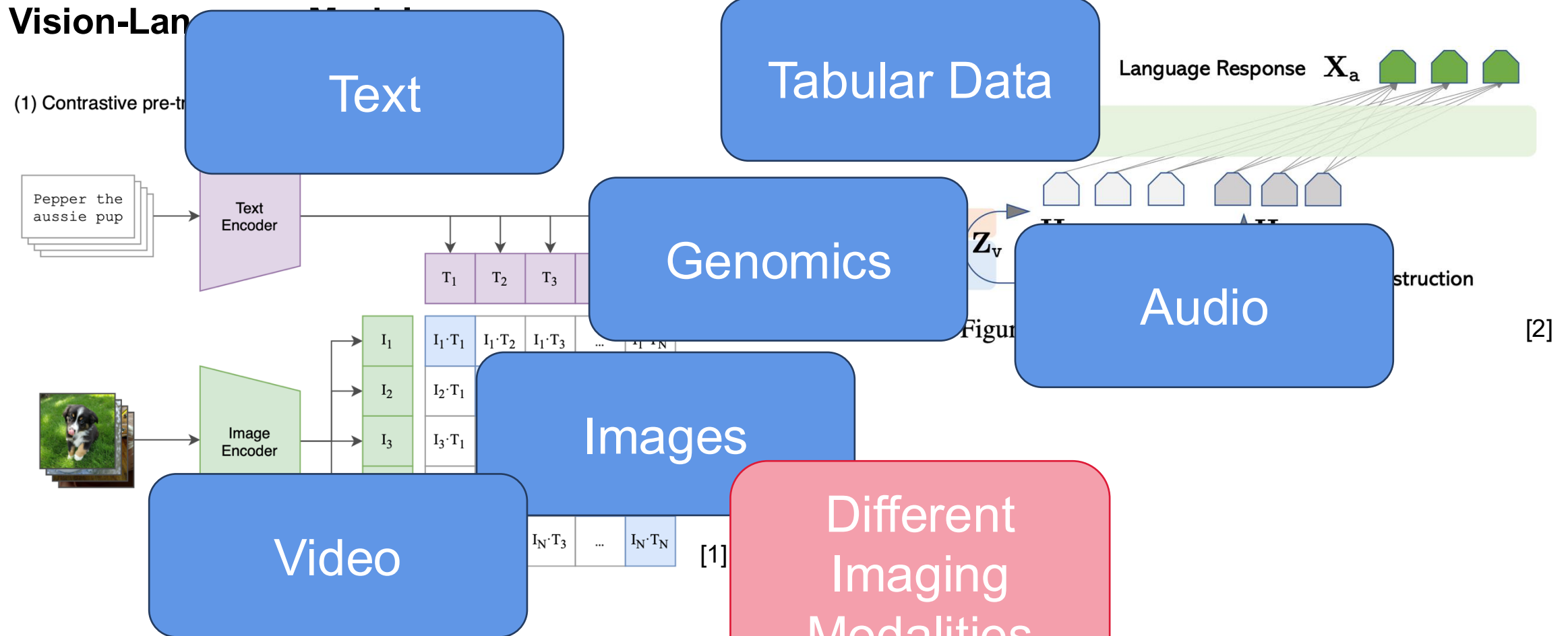
Figure 1: LLaVA network architecture.

[2]

[1] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." *International conference on machine learning*. Pmlr, 2021.

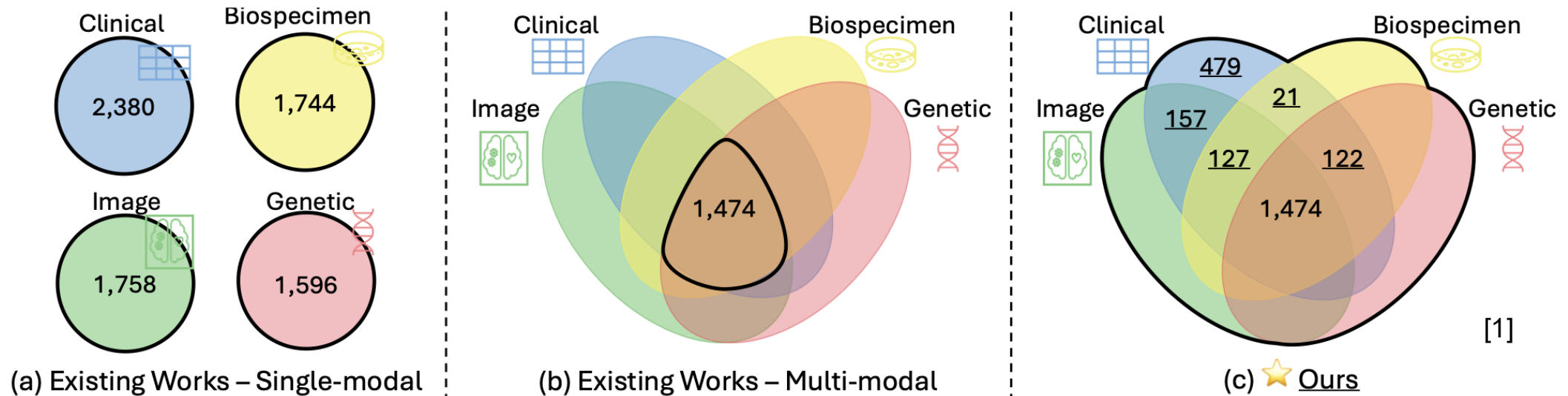
[2] Liu, Haotian, et al. "Visual instruction tuning." *Advances in neural information processing systems* 36 (2023): 34892-34916.

Multimodal Learning



- [1] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." *International conference on machine learning*. PMLR, 2021.
- [2] Liu, Haotian, et al. "Visual instruction tuning." *Advances in neural information processing systems* 36 (2023): 34892-34901.

Multimodal Learning



Multimodal methods often assume that all modalities are available

But in clinical practice that's not the case...

[1] Yun, Sukwon, et al. "Flex-moe: Modeling arbitrary modality combination via the flexible mixture-of-experts." Advances in Neural Information Processing Systems 37 (2024): 98782-98805.

Have a nice day!

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